



Large Scale Hybrid Computing

2026

André Pereira

Top HPC systems in TOP500 lists

(some images are borrowed)

What is TOP500?

TOP500

From Wikipedia, the free encyclopedia

The **TOP500** project ranks and details the 500 most powerful non-distributed computer systems in the world. The project was started in 1993 and publishes an updated list of the supercomputers twice a year. The first of these updates always coincides with the International Supercomputing Conference in June, and the second is presented at the ACM/IEEE Supercomputing Conference in November. The project aims to provide a reliable basis for tracking and detecting trends in high-performance computing and bases rankings on HPL,^[1] a portable implementation of the high-performance LINPACK benchmark written in Fortran for distributed-memory computers.

The 60th TOP500 was published in November 2022. Since June 2022, the United States' Frontier is the most powerful supercomputer on TOP500, reaching 1102 petaFlops (1.102 exaFlops) on the LINPACK benchmarks.^[2] The United States has by far the highest share of total computing power on the list (nearly 50%),^[3] while China currently leads the list in number of systems with 173 supercomputers, with the U.S. not far behind in second place.

The TOP500 list is compiled by Jack Dongarra of the University of Tennessee, Knoxville, Erich Strohmaier and Horst Simon of the National Energy Research Scientific Computing Center (NERSC) and Lawrence Berkeley National Laboratory (LBNL), and, until his death in 2014, Hans Meuer of the University of Mannheim, Germany.

APereira, LSHC, The TOP500 project lists also Green500 and HPCG benchmark list.

TOP500



The List.

Key people	Erich Strohmaier Jack Dongarra Horst Simon Martin Meuer
Established	24 June 1993
Website	www.top500.org

LINPACK benchmarks (HPL)

LINPACK benchmarks

From Wikipedia, the free encyclopedia

For the software library, see [LINPACK](#).

The **LINPACK Benchmarks** are a measure of a system's **floating point** computing power. Introduced by **Jack Dongarra**, they measure how fast a computer solves a dense n by n **system of linear equations** $Ax = b$, which is a common task in **engineering**.

The latest version of these **benchmarks** is used to build the **TOP500** list, ranking the world's most powerful supercomputers.^[1]

The aim is to approximate how fast a computer will perform when solving real problems. It is a simplification, since no single computational task can reflect the overall performance of a computer system. Nevertheless, the LINPACK benchmark performance can provide a good correction over the peak performance provided by the manufacturer. The peak performance is the maximal theoretical performance a computer can achieve, calculated as the machine's frequency, in cycles per second, times the number of operations per cycle it can perform. The actual performance will always be lower than the peak performance.^[2] The **performance of a computer** is a complex

HPL, written in C, measures the sustained floating-point rate (GFLOPs/s) to solve a dense system of linear equations using double-precision floating-point arithmetic

LINPACK benchmarks

Original author(s)	Jack Dongarra , Jim Bunch, Cleve Moler, and Gilbert Stewart
Initial release	1979
Website	www.netlib.org/benchmark/hpl/ 



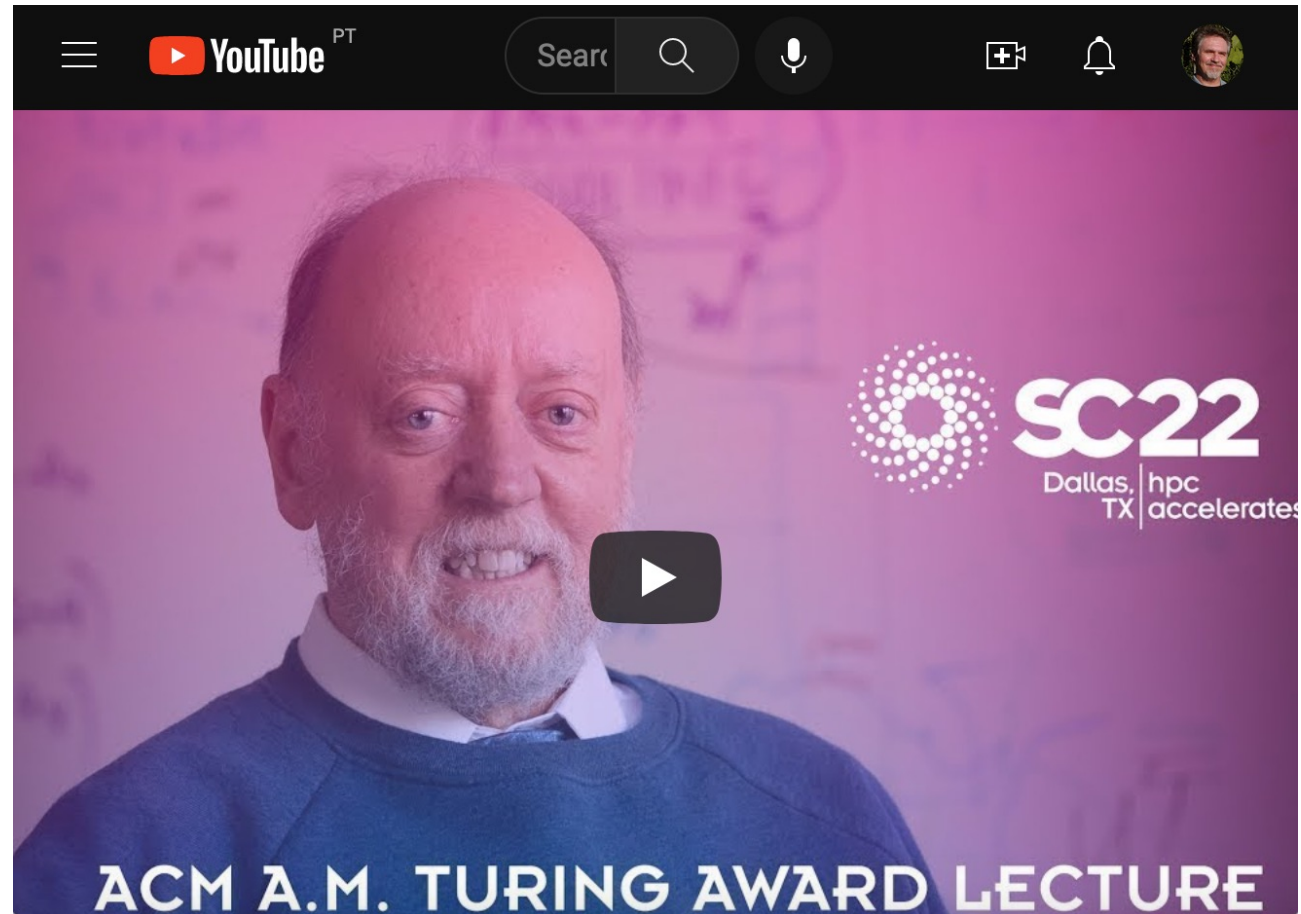
For pioneering contributions to numerical algorithms and libraries that enabled high performance computational software to keep pace with exponential hardware improvements

JACK DONGARRA



AWARD 2021
A.M. TURING

ACM Turing Award 2021: Jack Dongarra



<https://www.youtube.com/watch?v=lsnRP9akCDk>



1. **TOP500** (*LINPACK*)

- a) TOP10 lists from Nov'19 to Nov'23
- b) Analysis of some relevant systems and architectures
- c) Country distribution over the past 25 years
- d) PU chip technology evolution in the past 25 years & since last year
- e) Evolution of the accelerators since they were available

2. **GREEN500**

- a) TOP5 list from Nov'19 to Nov'22
- b) Analysis of some relevant systems and architectures

3. **HPCG**

- a) HPCG vs. HPL: an overview
- b) TOP10 in Nov'22

4. **HPL-AI**

- a) High-performance Linpack (HPL) and artificial intelligence (AI) workloads

Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,699,904	1,194.00	1,679.82	22,703
2	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	4,742,808	585.34	1,059.33	24,687
3	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Microsoft Azure United States	1,123,200	561.20	846.84	
4	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
5	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107
6	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy	1,824,768	238.70	304.47	7,404
7	Summit - IBM Power System AC922, IBM POWER9 22C 3.07GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM DOE/SC/Oak Ridge National Laboratory United States	2,414,592	48.60		
8	MareNostrum 5 ACC - BullSequana XH3000, Xeon Platinum 8460Y+ 40C 2.3GHz, NVIDIA H100 64GB, Infiniband NDR200, EVIDEN EuroHPC/BSC Spain	680,960	138.20		
9	Eos NVIDIA DGX SuperPOD - NVIDIA DGX H100, Xeon Platinum 8480C 56C 3.8GHz, NVIDIA H100, Infiniband NDR400, Nvidia NVIDIA Corporation United States			485,888	121.40
10	Sierra - IBM Power System AC922, IBM POWER9 22C 3.1GHz, NVIDIA Volta GV100, Dual-rail Mellanox EDR Infiniband, IBM / NVIDIA / Mellanox DOE/NNSA/LLNL United States			1,572,480	94.64

Since June 22

Top 10 HPC systems Nov'23 TOP500

New in the TOP10 list,
with 4th Gen Xeon & Intel GPU

New in the TOP10 list,
with 4th Gen Xeon & H100

3 EuroHPC systems in
the TOP10 list!

Top 10 HPC systems Nov'25 TOP500

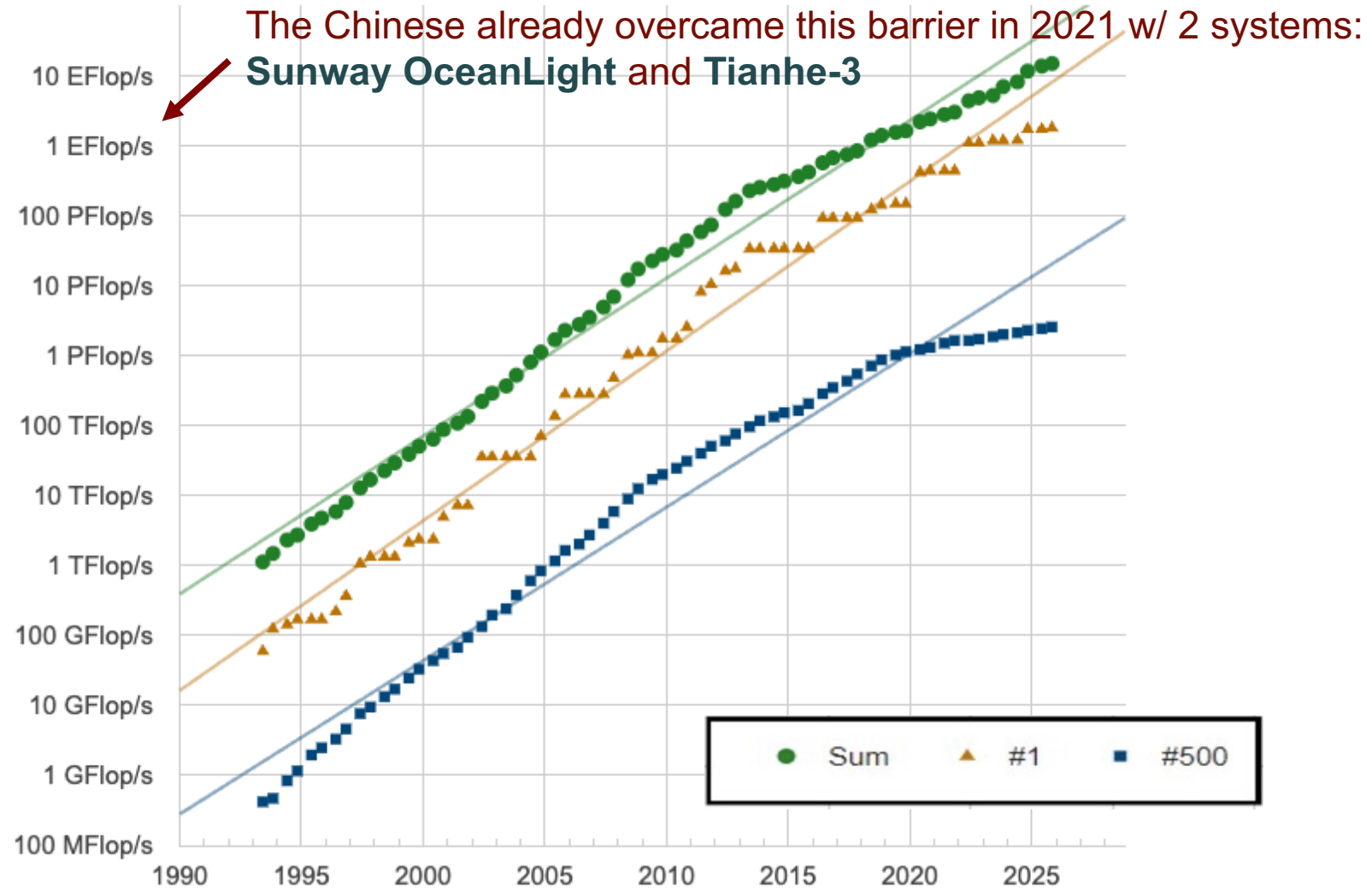
Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	2,821.10	29,685
2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,760.01	38,698
4	JUPITER Booster - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN EuroHPC/FZJ Germany	4,801,344	1,000.00	1,226.28	15,794
5	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	
6	HPC6 - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.9, HPE Eni S.p.A. Italy	3,143,520	477.90	606.97	8,461
7	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
8	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	2,121,600	434.90	574.84	7,124
9	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107
10	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy	1,824,768	241.20	306.31	7,494



New in the TOP10 list,
with 4th Gen Xeon & Intel GPU

Nvidia losing market share

5 European systems in
the TOP10 list!



Analysis of some key systems in 2025



1. **#2 Frontier** (*AMD Epyc Trento 64c + Instinct MI250x*)
2. **#10 Leonardo**, (*3rd Gen Xeon, 32c + NVidia Ampere A100*)
3. **#5 Jupiter** (*72c Nvidia GH200 superchip*)
4. **#7** (*#1 in Jun'20*): **Fugaku** (*Fujitsu A64FX, 48 cores*)



Frontier at ORNL



Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE
DOE/SC/Oak Ridge National Laboratory
United States
Nov'23

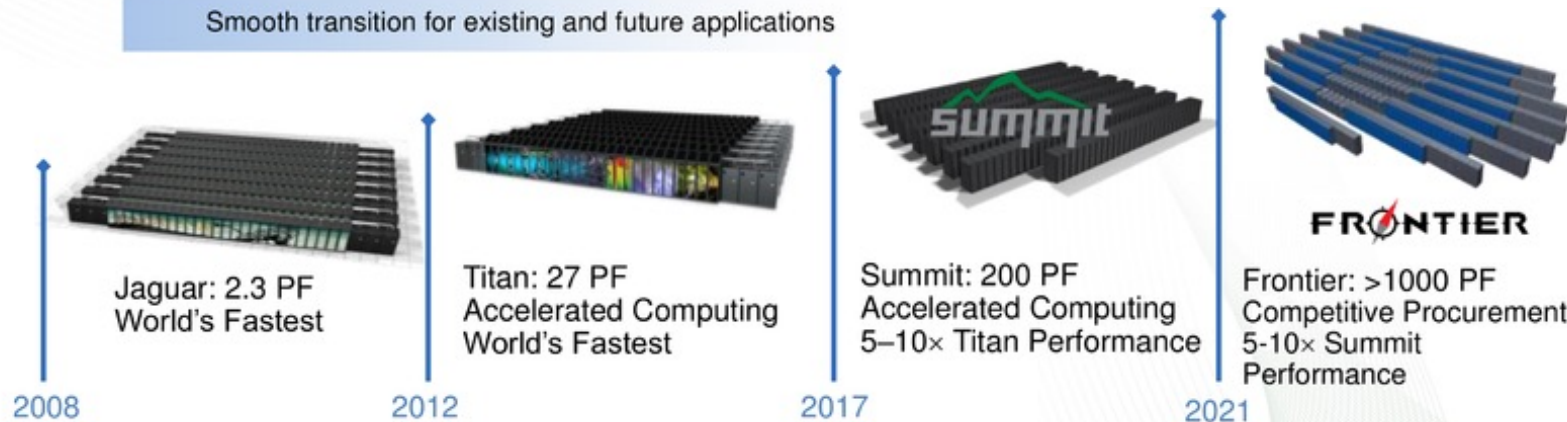
Competitive procurement asking for:

50–100× application performance of Titan

Support for traditional modeling and simulation, high-performance data analysis, and artificial intelligence applications

Peak performance >1000 PF

Smooth transition for existing and future applications



JAGUAR

No. 1 in 2009, 2010



TITAN

No. 1 in 2012



SUMMIT

No. 1 in 2018



FRONTIER

No. 1 in 2022

Frontier Overview Extraordinary Engineering



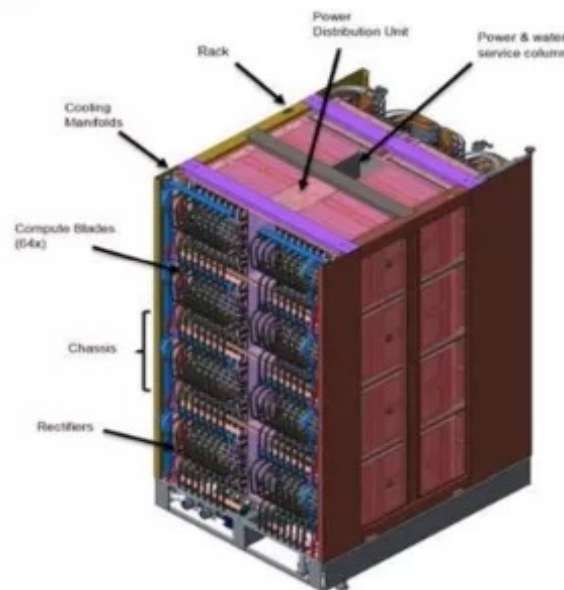
System

- 2 EF Peak DP FLOPS
- 74 compute racks
- 29 MW Power Consumption
- 9,408 nodes
- 9.2 PB memory (4.6 PB HBM, 4.6 PB DDR4)
- Cray Slingshot network with dragonfly topology
- 37 PB Node Local Storage
- 716 PB Center-wide storage
- 4000 ft² foot print

Built by HPE

Olympus rack

- 128 AMD nodes
- 8,000 lbs
- Supports 400 KW



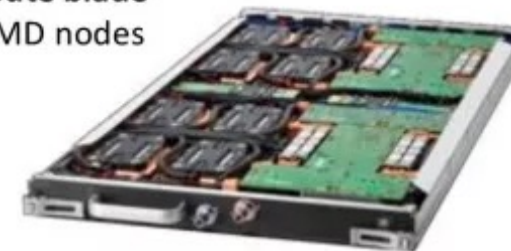
Powered by AMD

AMD node

- 1 AMD "Trento" CPU
- 4 AMD MI250X GPUs
- 512 GiB DDR4 memory on CPU
- 512 GiB HBM2e total per node (128 GiB HBM per GPU)
- Coherent memory across the node
- 4 TB NVM
- GPUs & CPU fully connected with AMD Infinity Fabric
- 4 Cassini NICs, 100 GB/s network BW

Compute blade

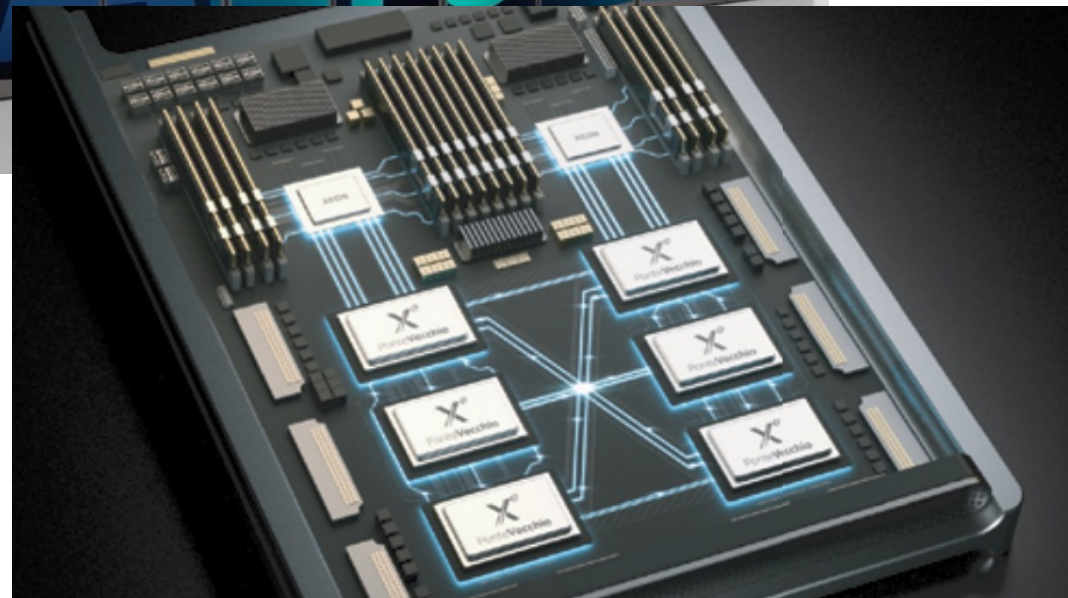
- 2 AMD nodes



All water cooled, even DIMMS and NICs



> 2 EFlops
38 MW !!!!
500 M USD





System Overview

Intel-Hewlett Packard
Enterprise (HPE)
System

Exascale System

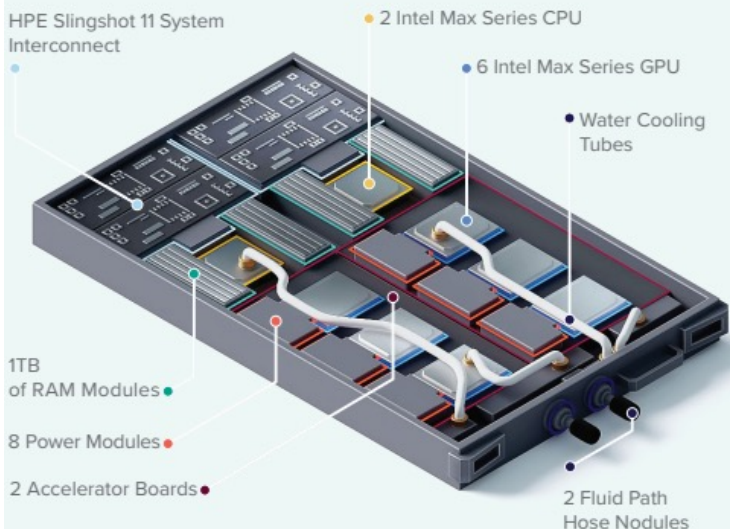
10,624 Compute
Blades

21,248 Intel Xeon
CPU Max Series

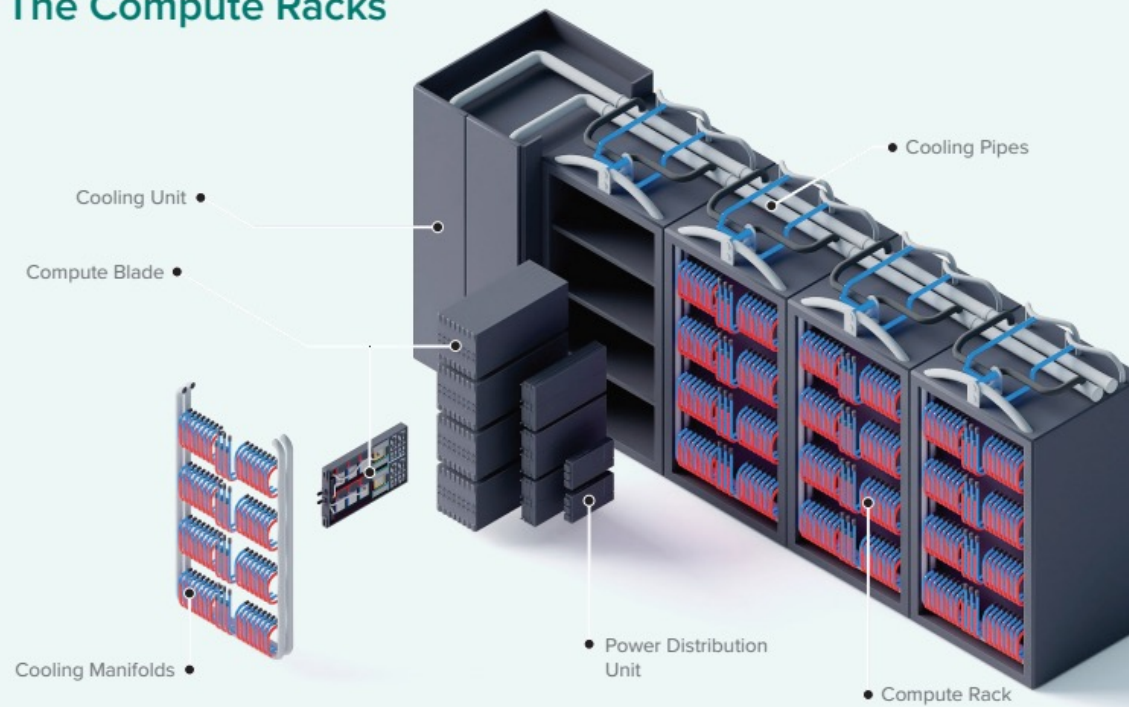
63,744 Intel Data
Center GPU Max
Series

166 Racks

The Compute Blade

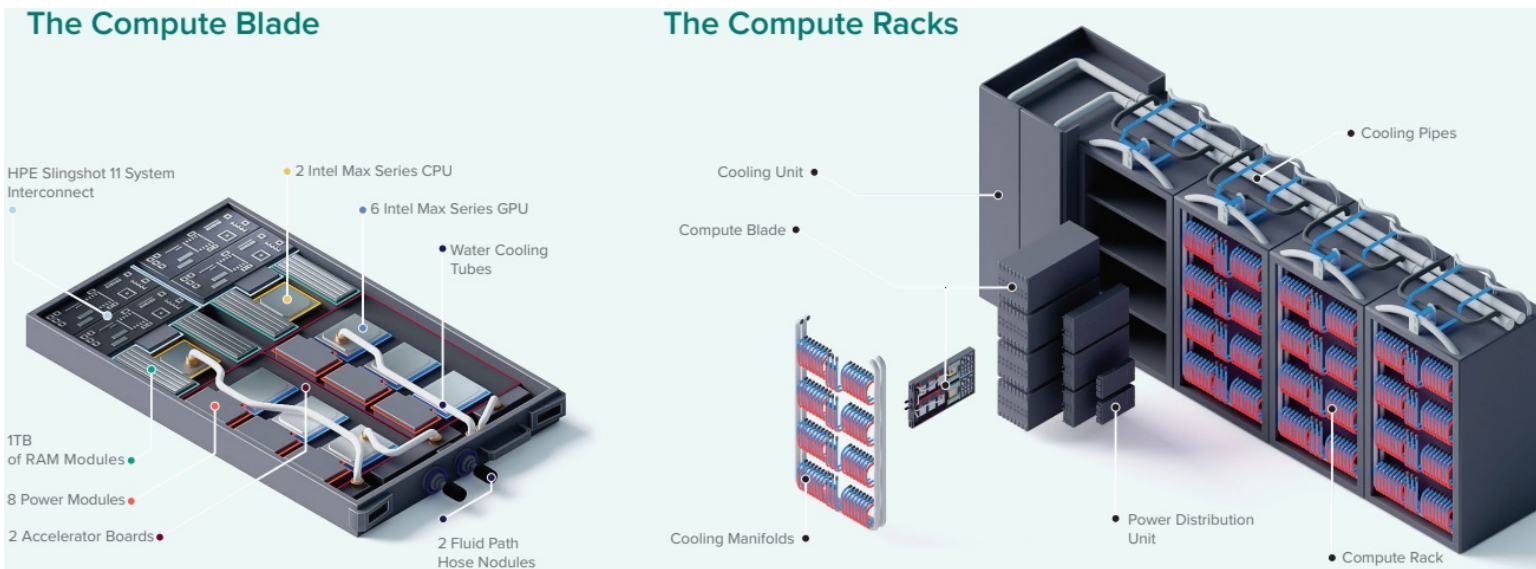


The Compute Racks





Aurora at ANL



Compute Node

2 Intel Xeon CPU Max Series processors; 6 Intel Data Center Max GPUs; Unified Memory Architecture; 8 fabric endpoints; RAMBO

GPU Architecture

Intel Data Center GPU Max Series; Tile-based, chiplets, HBM stack, Foveros 3D integration

On-Node Interconnect

CPU-GPU: PCIe
GPU-GPU: Xe Link

Aggregate System Memory

20.4 PB

High-Performance Storage

230 PB, 31 TB/s (DAOS)

System Interconnect

HPE Slingshot 11; Dragonfly topology with adaptive routing

Network Switch

25.6 Tb/s per switch, from 64–200 Gbs ports (25 GB/s per direction)

Programming Environment

Intel oneAPI, MPI, OpenMP, C/C++, Fortran, SYCL/DPC++

Software Stack

HPE Cray EX software stack + Intel Enhancements + Data and Learning

Platform

HPE Cray EX supercomputer

System Performance

Exascale

System Size

10,624 nodes

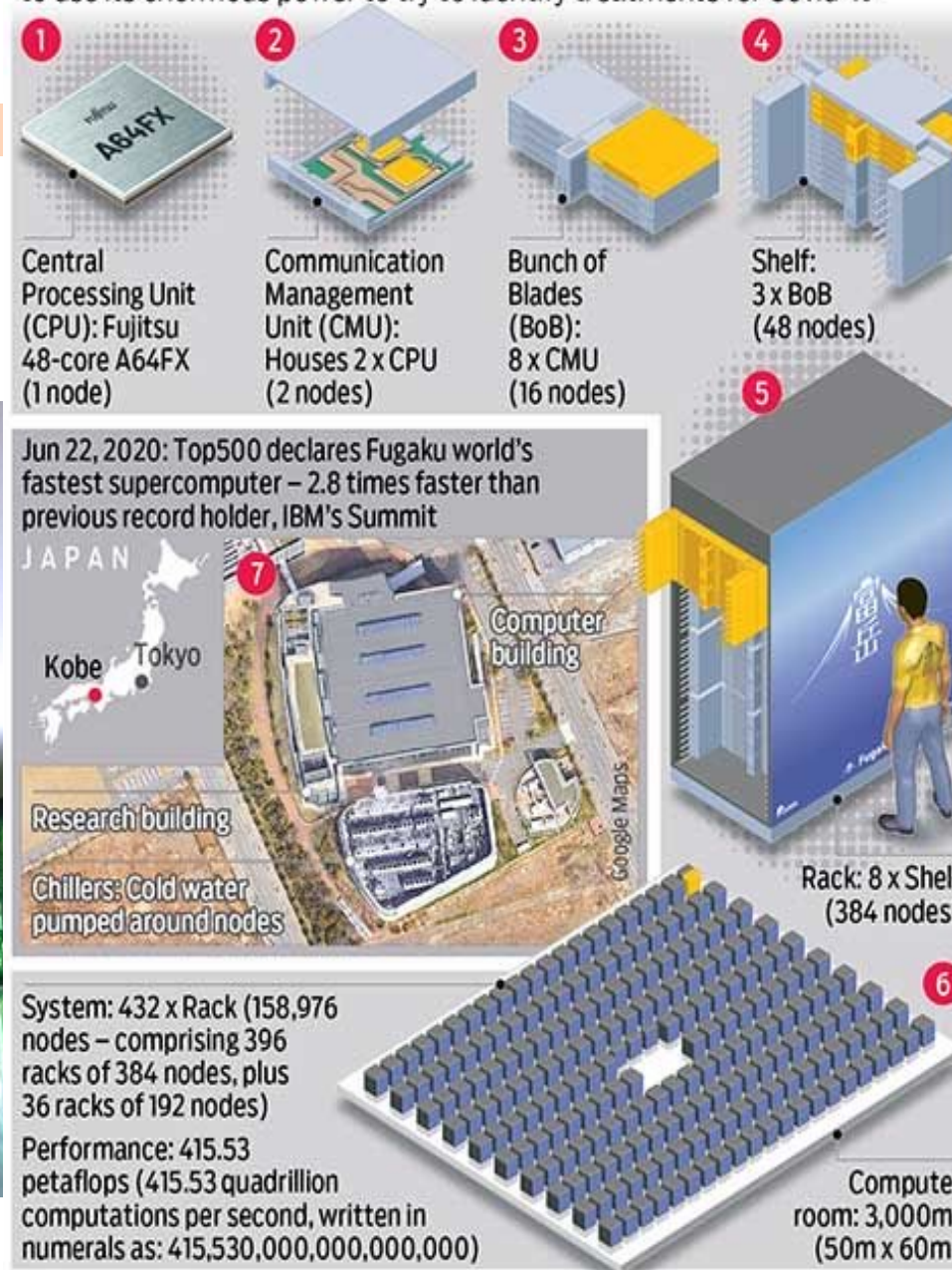
②

富士

Fugaku

Supercomputer to seek Covid-19 cure

The world's fastest supercomputer, Japan's \$1.2 billion Fugaku, is to use its enormous power to try to identify treatments for Covid-19



4 Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan **Nov'23**

TOP 500
The List.

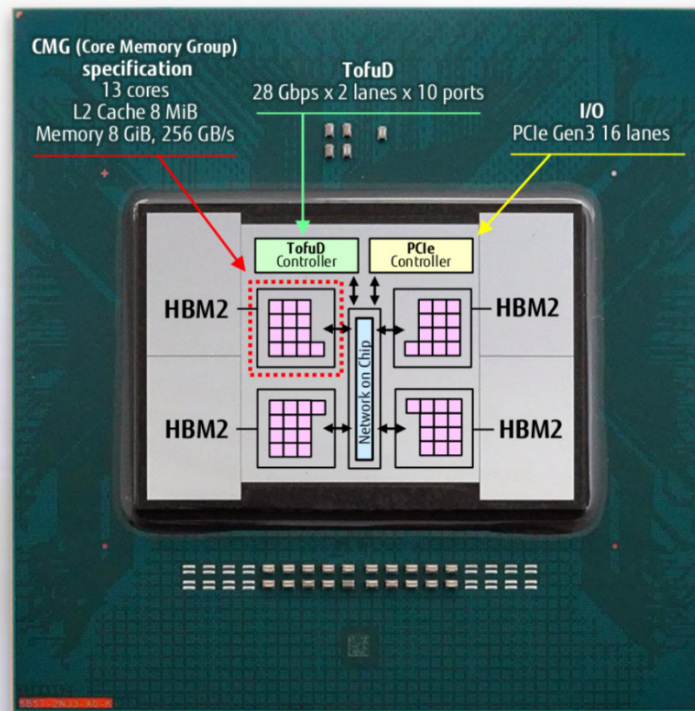




Fujitsu A64FX in Fugaku



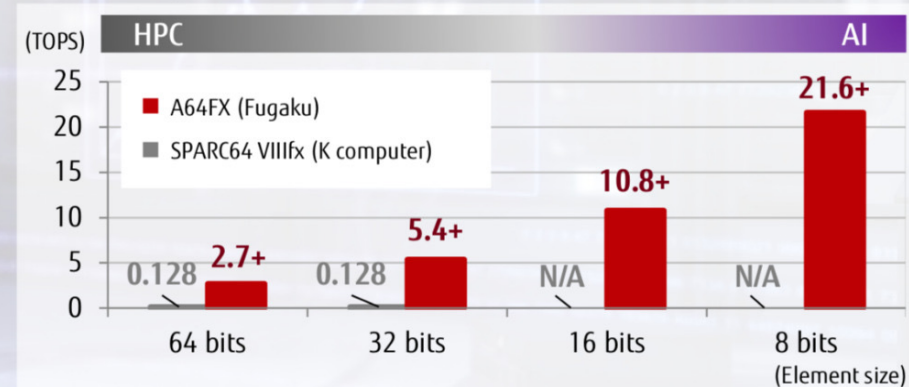
1. High-Performance Arm CPU A64FX in HPC and AI Areas



Architecture features

ISA	Armv8.2-A (AArch64 only) SVE (Scalable Vector Extension)	arm
SIMD width	512-bit	
Precision	FP64/32/16, INT64/32/16/8	
Cores	48 computing cores + 4 assistant cores (4 CMGs)	
Memory	HBM2: Peak B/W 1,024 GB/s	
Interconnect	TofuD: 28 Gbps x 2 lanes x 10 ports	

Peak performance (Chip level)



EuroHPC Supercomputers at the TOP10

TOP 500 Nov'23
The List.



EuroHPC
Joint Undertaking

3 EuroHPC
SUPERCOMPUTERS
RANKED AMONG THE
WORLD'S TOP 10
SUPERCOMPUTERS

TOP 500

LUMI #5

LEONARDO #6

MARENOSTRUM 5 #8





LUMI at CSC, Finland



1 SYSTEM
380 Petaflop/s
PEAK PERFORMANCE

COMPUTING POWER EQUALS
1.5 MILLION
MODERN LAPTOP'S
CAPACITY

100%
HYDROELECTRIC
ENERGY

LUMI-C:
x86 Partition
Supplementary CPU partition:
over **262,000**
AMD EPYC CPU cores.

LUMI-K:
Container Cloud Service

LUMI-O:
Object Storage Service
30 PB
encrypted object storage
(Ceph) for storing, sharing
and staging data.

LUMI-Q:
Quantum Computing

High-speed interconnect

Possibility for combining
different resources within
a single run. HPE
Slingshot technology.

LUMI-G:
GPU Partition
Sustained performance
380
Pflop/s powered by AMD
Radeon Instinct™ MI250X GPUs.

LUMI-D:
Data Analytics Partition
Interactive partition with
32 TB
of memory and graphics GPUs for
data analytics and visualization.

LUMI-F:
Accelerated Storage
10 PB
Flash-based storage layer with
extreme I/O bandwidth of
2 TB/s and IOPS capability.

LUMI-P:
Lustre Storage
80 PB
parallel file system.

<https://leonardo-supercomputer.cineca.eu/hpc-system>



**Barcelona
Supercomputing
Center**

Centro Nacional de Supercomputación

MareNostrum 5 at BSC, Spain



1120 nodes

- 2x Intel Sapphire Rapids 8460Y+ 40c each (80 cores node)
- 512 GB of Main memory, using DDR5
- 4x Nvidia Hopper GPU with 64 HBM2 memory
- 480GB on NVMe storage (/scratch)
- 4x NDR200 (BW per node 800Gb/s)

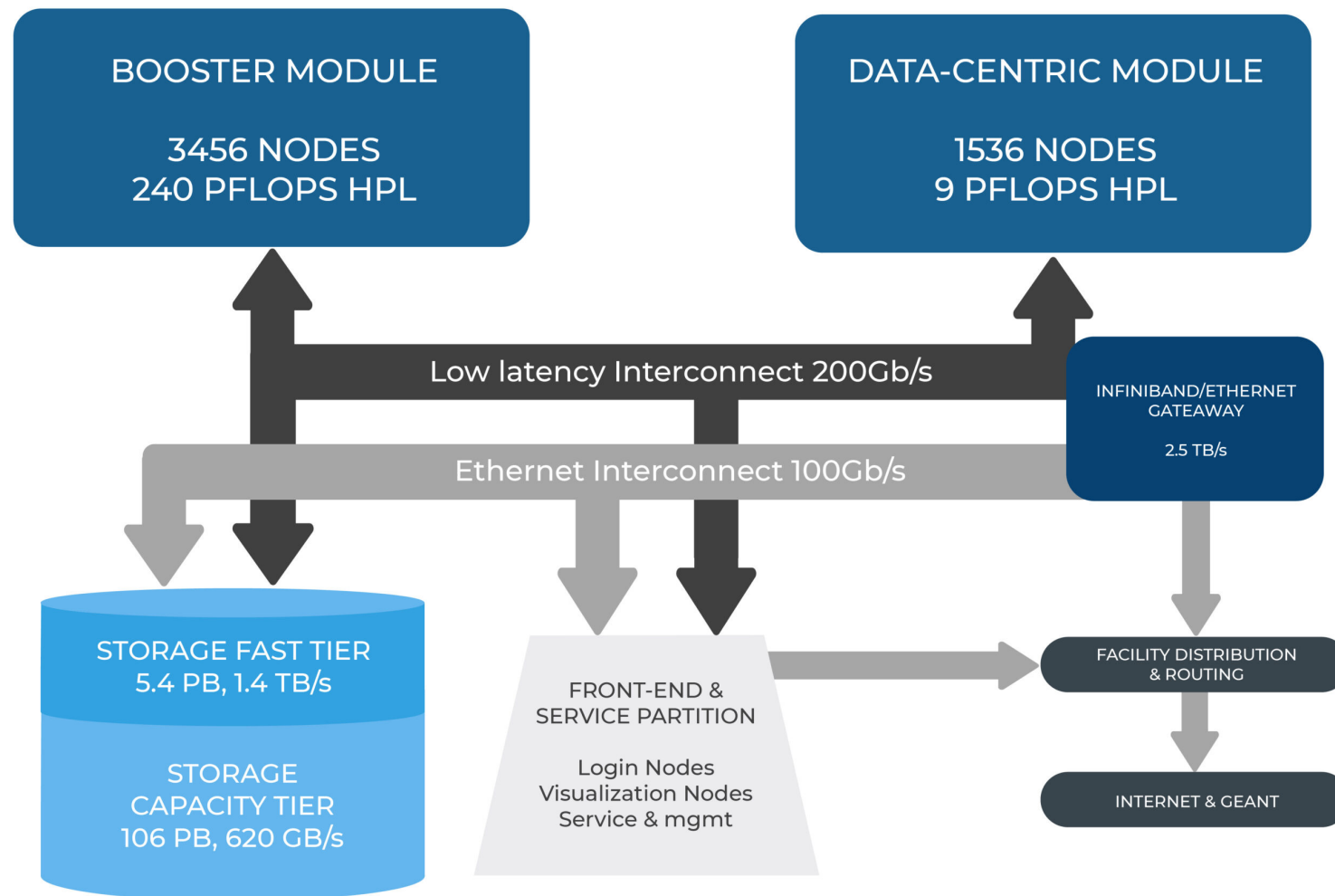
6500 nodes

- 2x Intel Sapphire Rapids 8480+ 56c each (112 cores per node)
- 256 GB DDR5 (with 216 nodes with 1024GB)
- 960GB on NVMe storage (/scratch)
- 1x NDR200 shared by 2 nodes (SharedIO) (BW per node 100Gb/s)





Leonardo at CINECA, Italy



<https://leonardo-supercomputer.cineca.eu/hpc-system/>



Leonardo at CINECA, Italy



BOOSTER MODULE

3456 nodes

Features a **custom BullSequana X2135 “Da Vinci” blade**, composed of:

- 1 x CPU Intel Xeon 8358 32 cores, 2.6 GHz
- 512 (8 x 64) GB RAM DDR4 3200 MHz
- 4 x NVidia custom Ampere GPU 64GB HBM2
- 2 x NVidia HDR 2x100 Gb/s cards

Performance per node: 89,4 TFLOPs peak



6

Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy



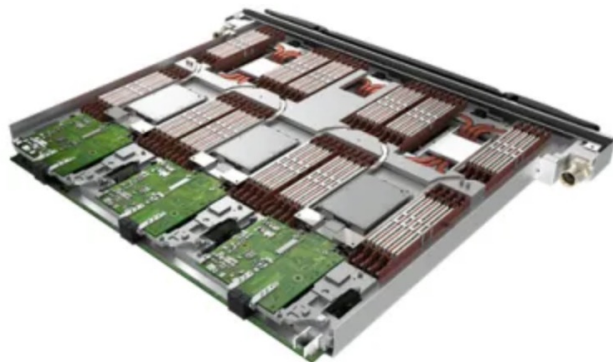
DATA CENTRIC MODULE

1536 nodes

Features a **BullSequana X2610 compute blade**.

Each computing node is composed of:

- 2x Intel Sapphire Rapids, 56 cores, TDP 350 W
- 512 (16 x 32) GB RAM DDR5 4800 MHz
- 1 x NVidia HDR100 100 Gb/s card
- 8 TB NVM





PAGE CONTENTS

LUMI

LEONARDO

MARENOSTRUM 5

MELUXINA

KAROLINA

DISCOVERER

VEGA

DEUCALION

JUPITER

in TOP10

DEUCALION

Deucalion is a petascale EuroHPC supercomputer located in Guimarães, Portugal. It is supplied by Fujitsu Technology Solutions combining a Fujitsu PRIMEHPC (ARM partition) and Atos Bull Sequana (x86 partitions). Deucalion is hosted by [MACC](#).



FCT

7.22 petaflops

Sustained performance

10.00 petaflops

Peak performance

Compute partitions:

ARM Partition: 1632 nodes, 3.8 PFLOps ; x86 Partition: 500 nodes, 1,62 PFLOps ; Accelerated: 33 nodes, 1,72 PFLOps

Central Processing Unit (CPU):

A64FX (ARM partition), AMD EPYC (x86 partitions)

Graphics Processing Unit (GPU):

NVidia Ampere

Storage capacity:

430 TB High-speed NVMe partition, 10.6 PB high-speed based Parallel File System partition.

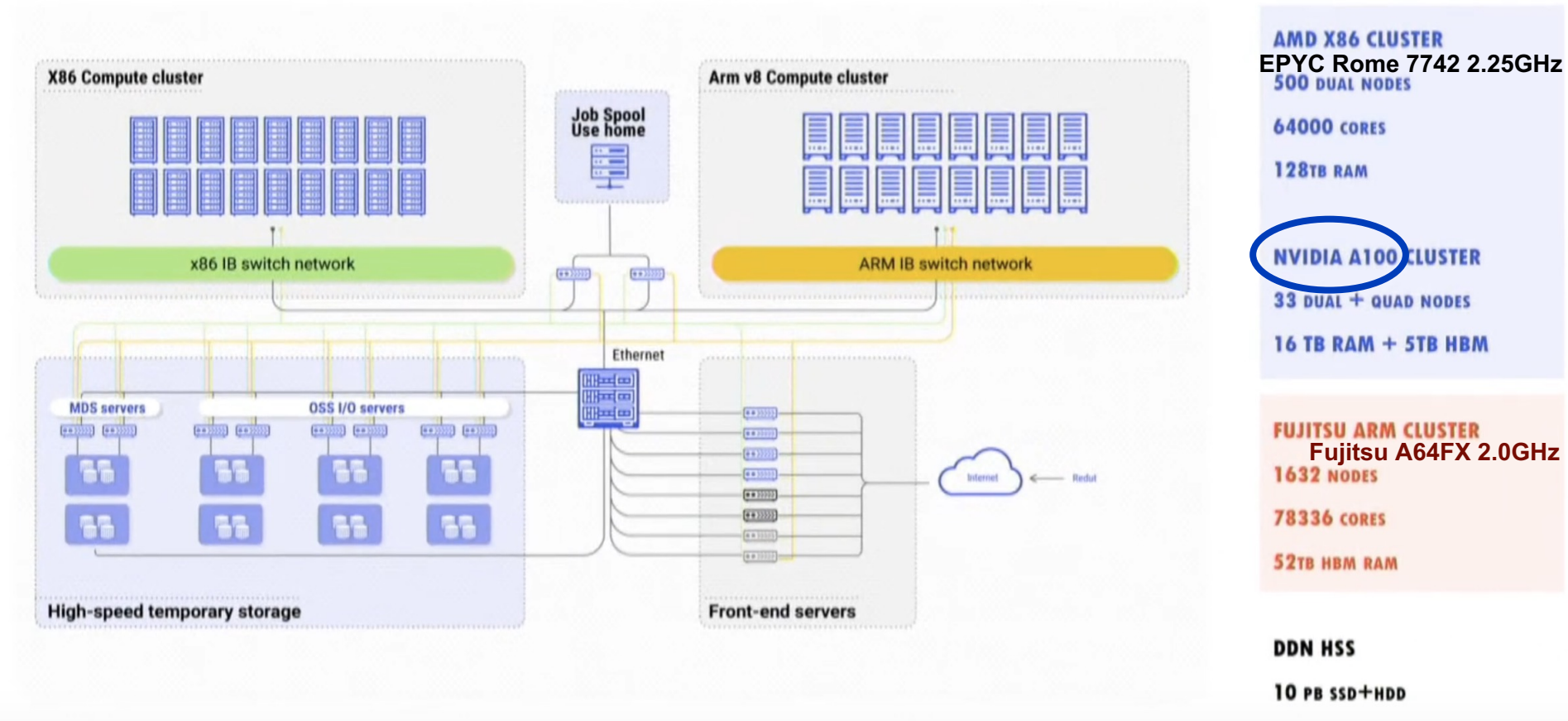
Applications:

Traditional Computational, AI, Big Data

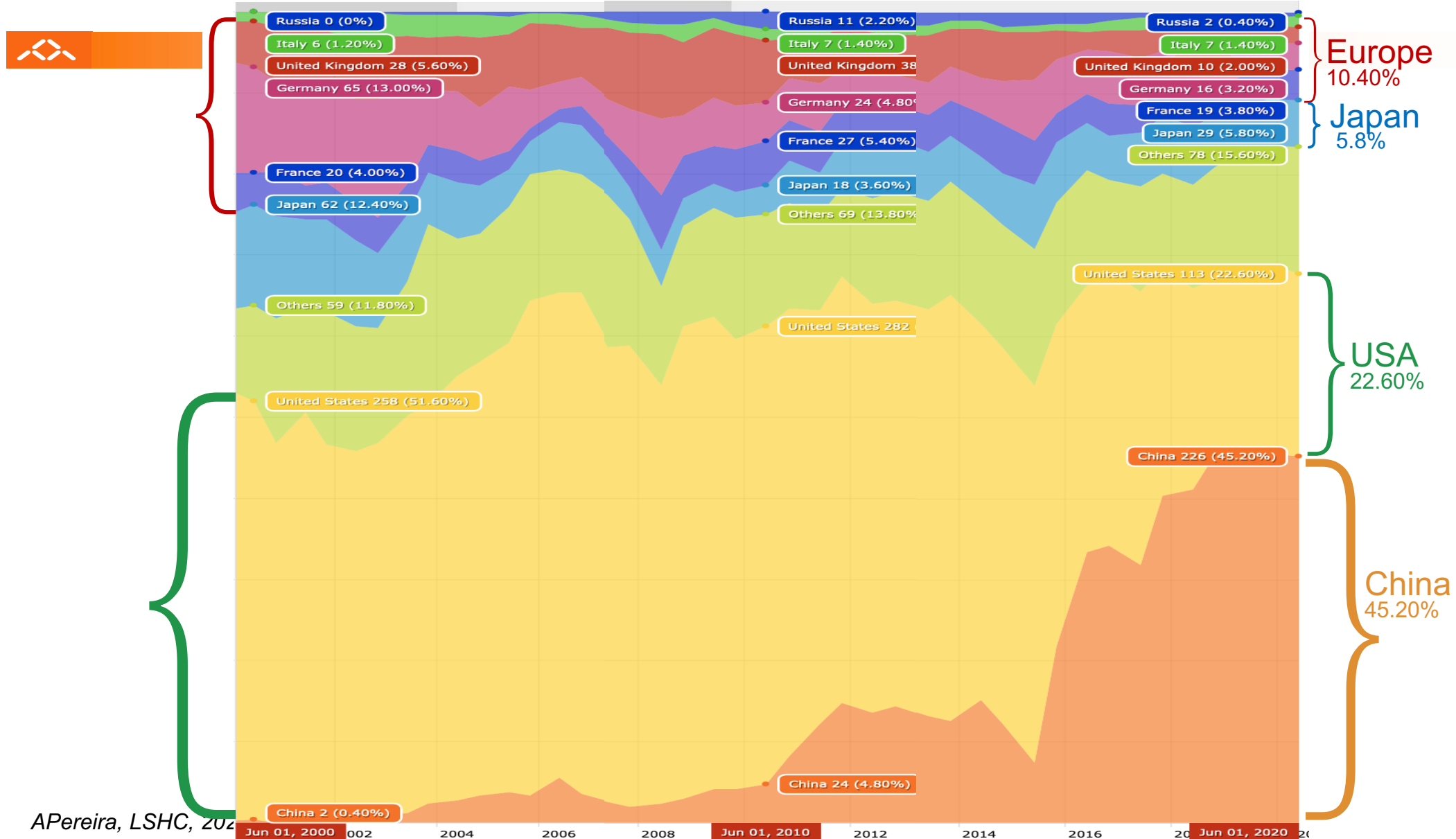
The Deucalion in MACC



DEUCALION OVERALL ARCHITECTURE

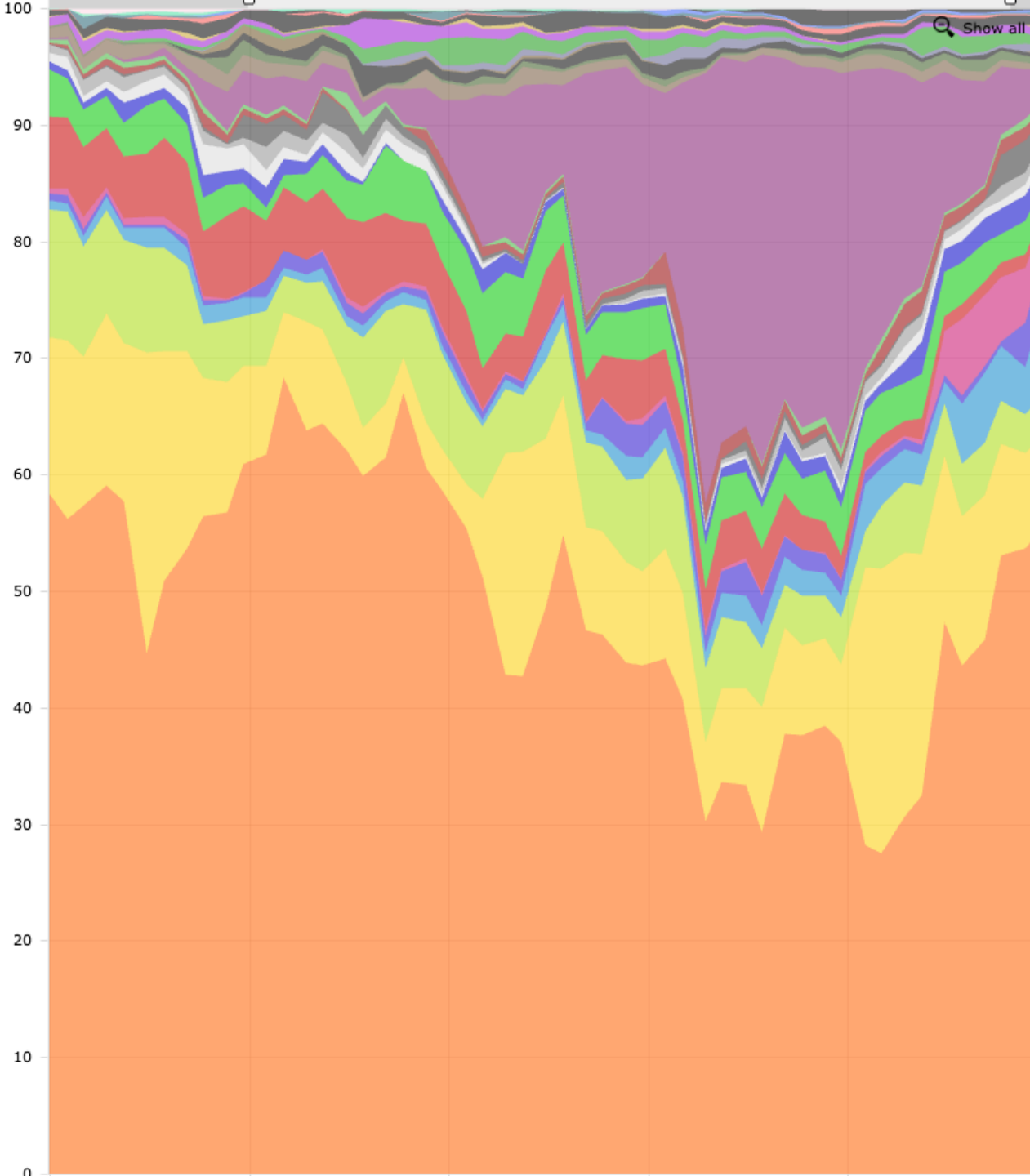


Country distribution over 20 years: from 2000 to 2020, #systems

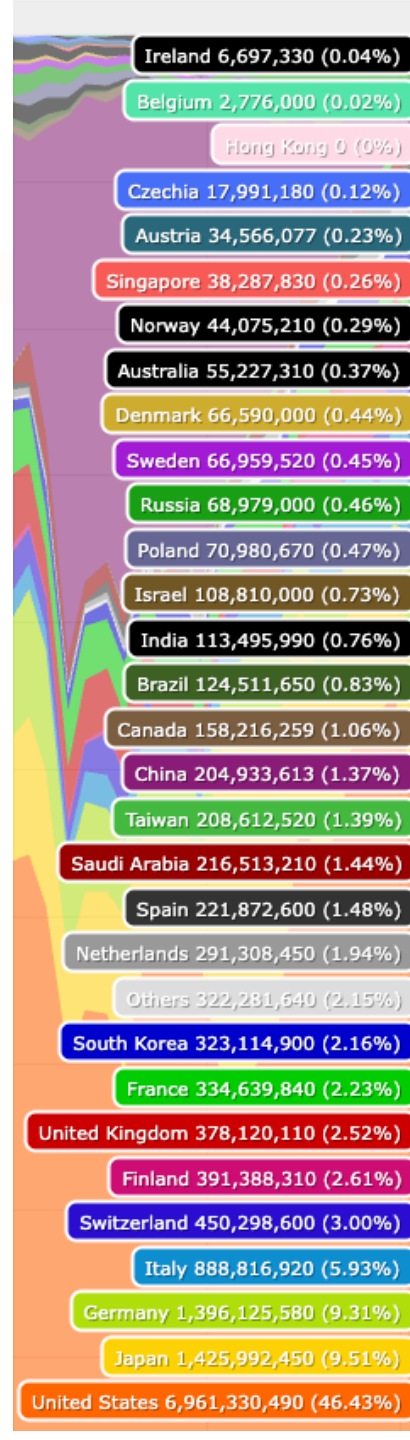




TOP



over 20 years:
performance

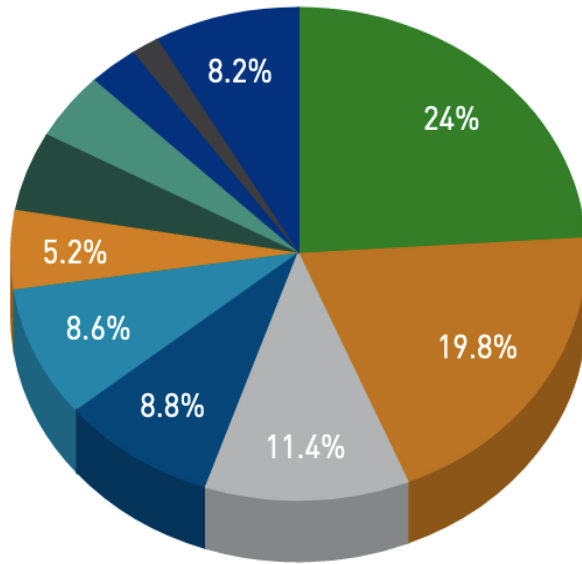




Processor generation share Nov'22

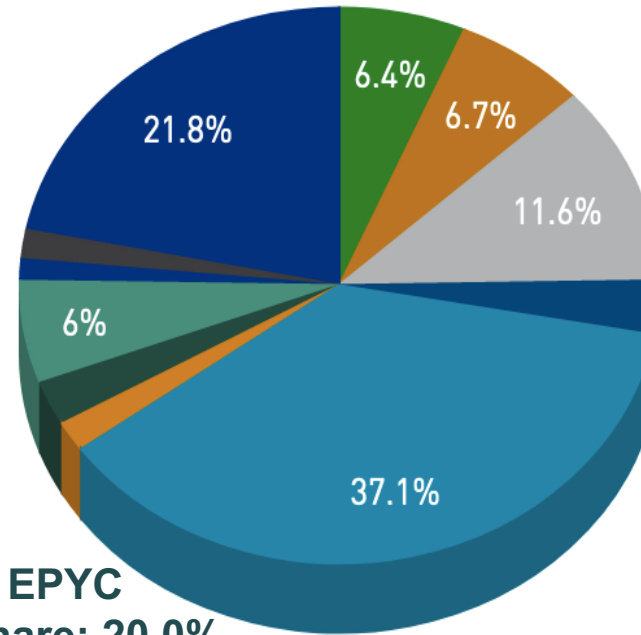


Processor Generation System Share



Processor Generation Performance Share

Intel Xeon
System share: 70.5%
Performance share: 27.8%

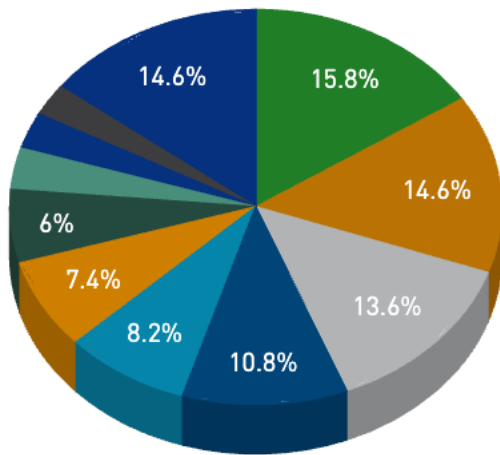


AMD EPYC
System share: 20.0%
Performance share: 48.7%

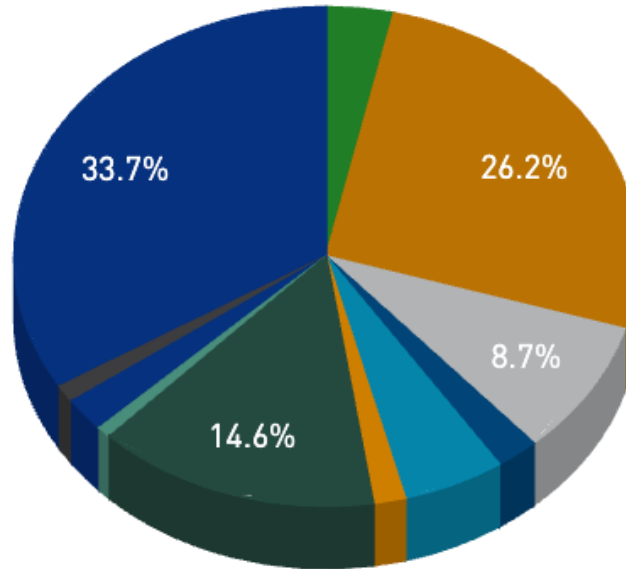
- Xeon Gold
- Xeon Gold 62xx (Cascade Lake)
- AMD Rome
- Xeon Platinum 82xx (...)
- AMD Milan
- Intel Xeon E5 (Broad...)
- Xeon Platinum
- Xeon® Platinum 83x...
- Intel Xeon E5 (Hasw...)
- Intel Xeon Phi
- Others



Processor Generation System Share



Processor Generation Performance Share

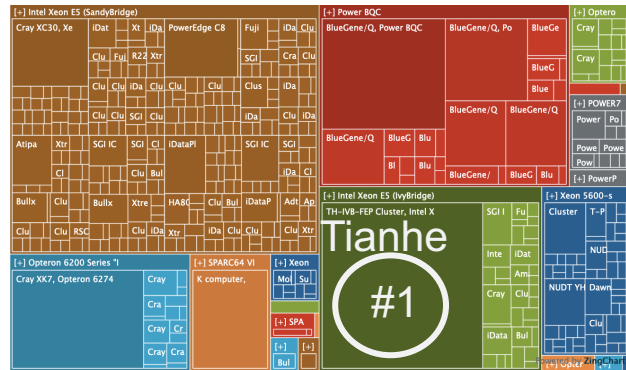


- Xeon Gold 62xx (Cascade Lake)
- AMD Zen-3 (Milan)
- AMD Zen-2 (Rome)
- Xeon Gold (Skylake)
- Xeon Platinum 83xx (Ice Lake)
- Xeon Platinum 82xx (Cascade Lake)
- Xeon Platinum (Sapphire Rapids)
- Intel Xeon E5 (Broadwell)
- AMD Zen-4 (Genoa)
- Xeon Platinum (Skylake)
- Others

Architecture family of key #1's: Nov'13 to Nov'23

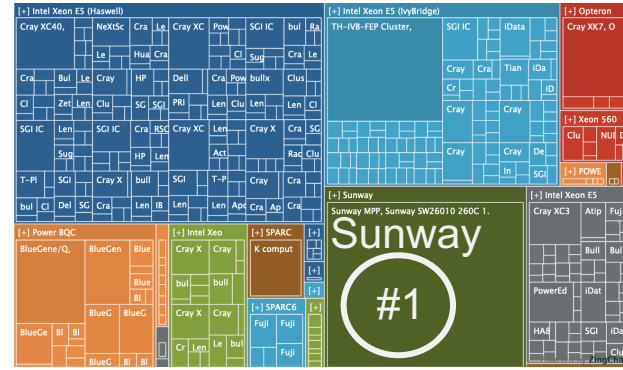


Nov'13



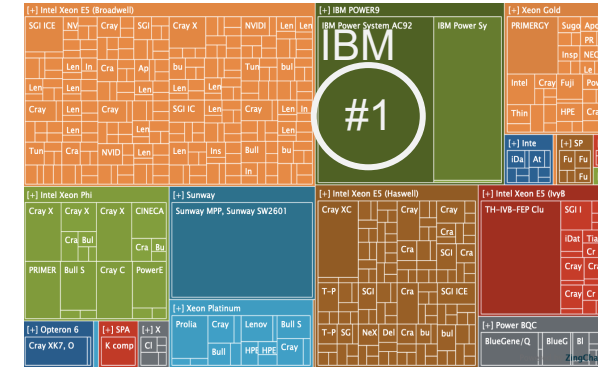
with Xeon Phi

Jun'16



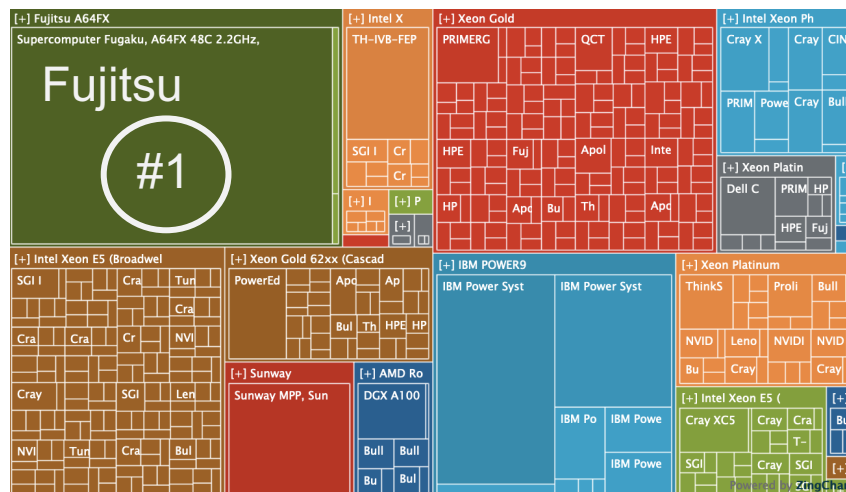
without any accelerator

Jun'18



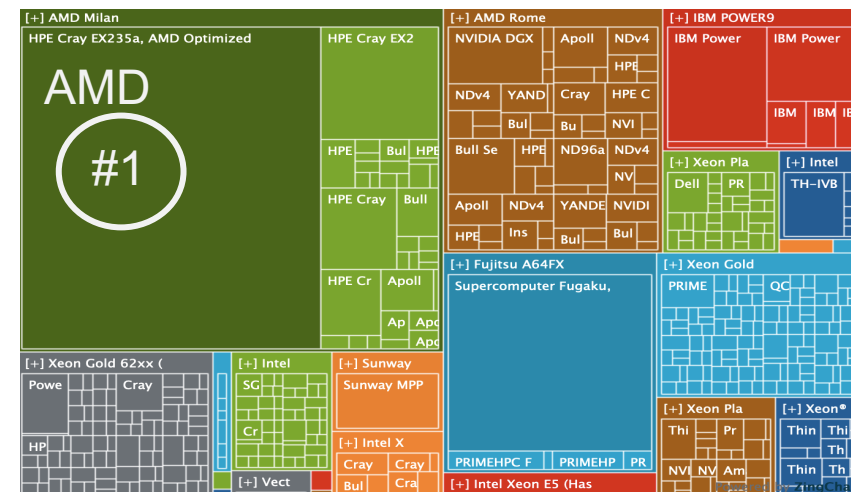
with Nvidia Volta

Jun'20



without any accelerator

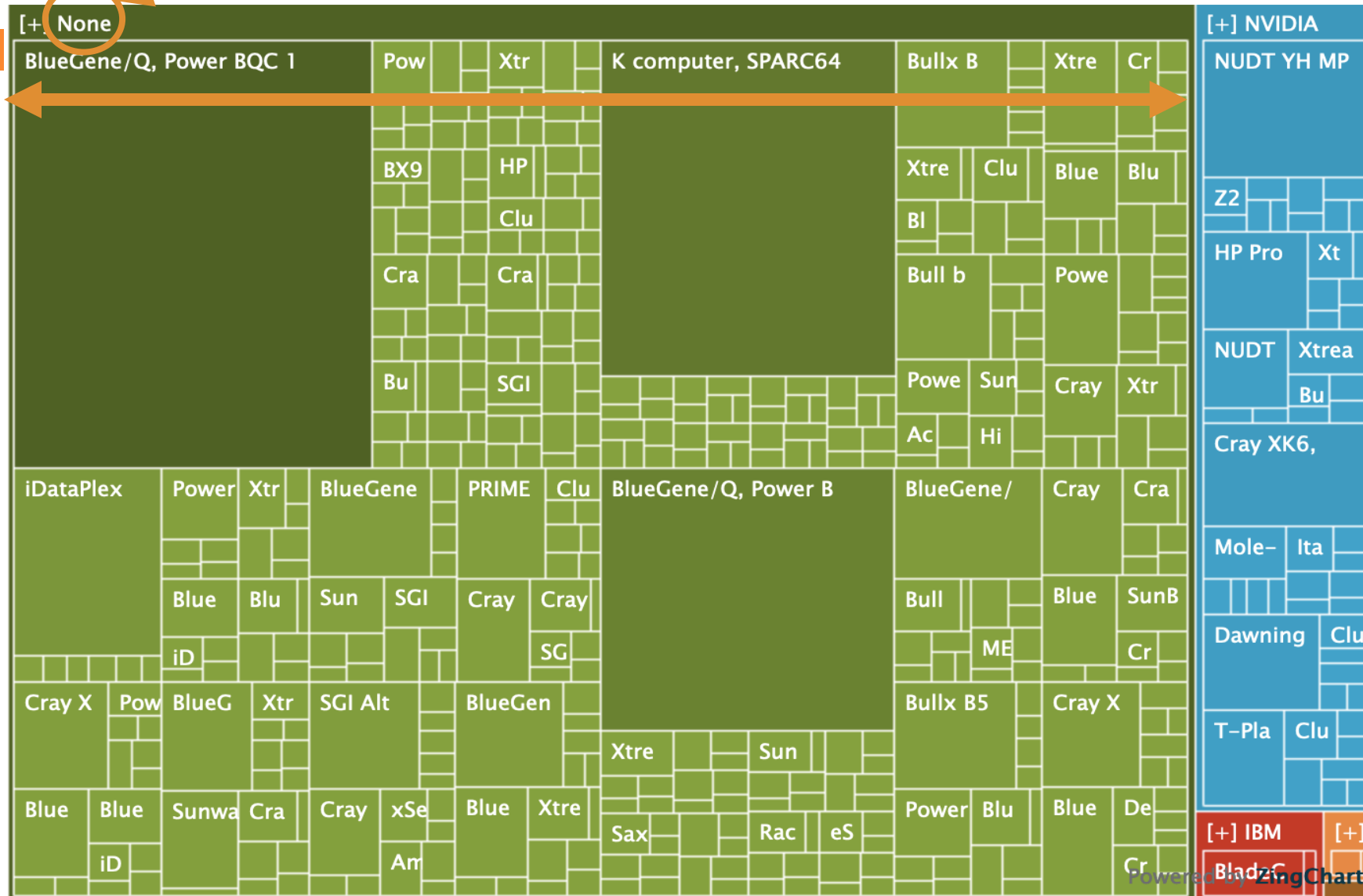
Jun'22



with AMD MI250x

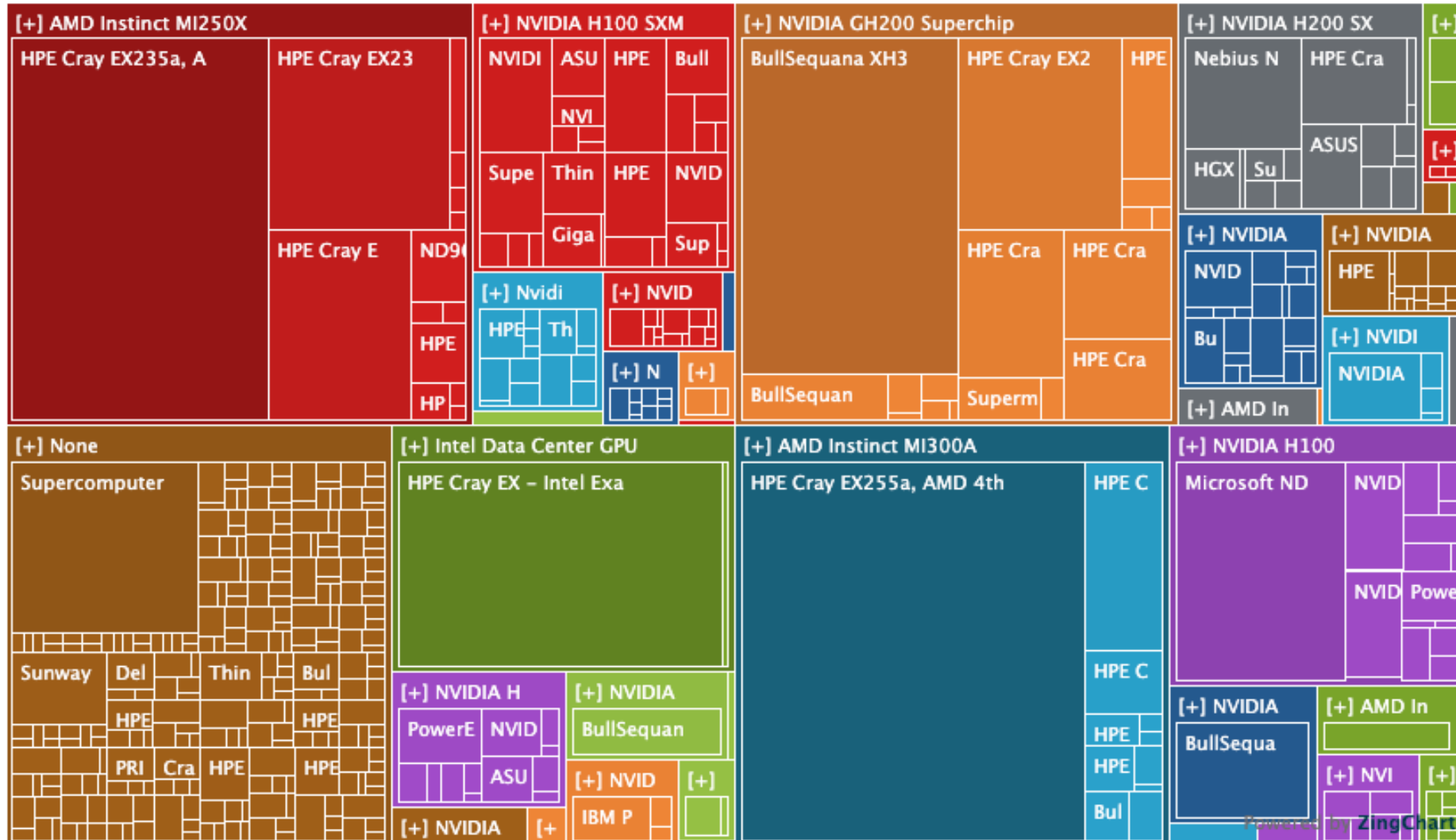
Accelerator family distribution

Jun'12

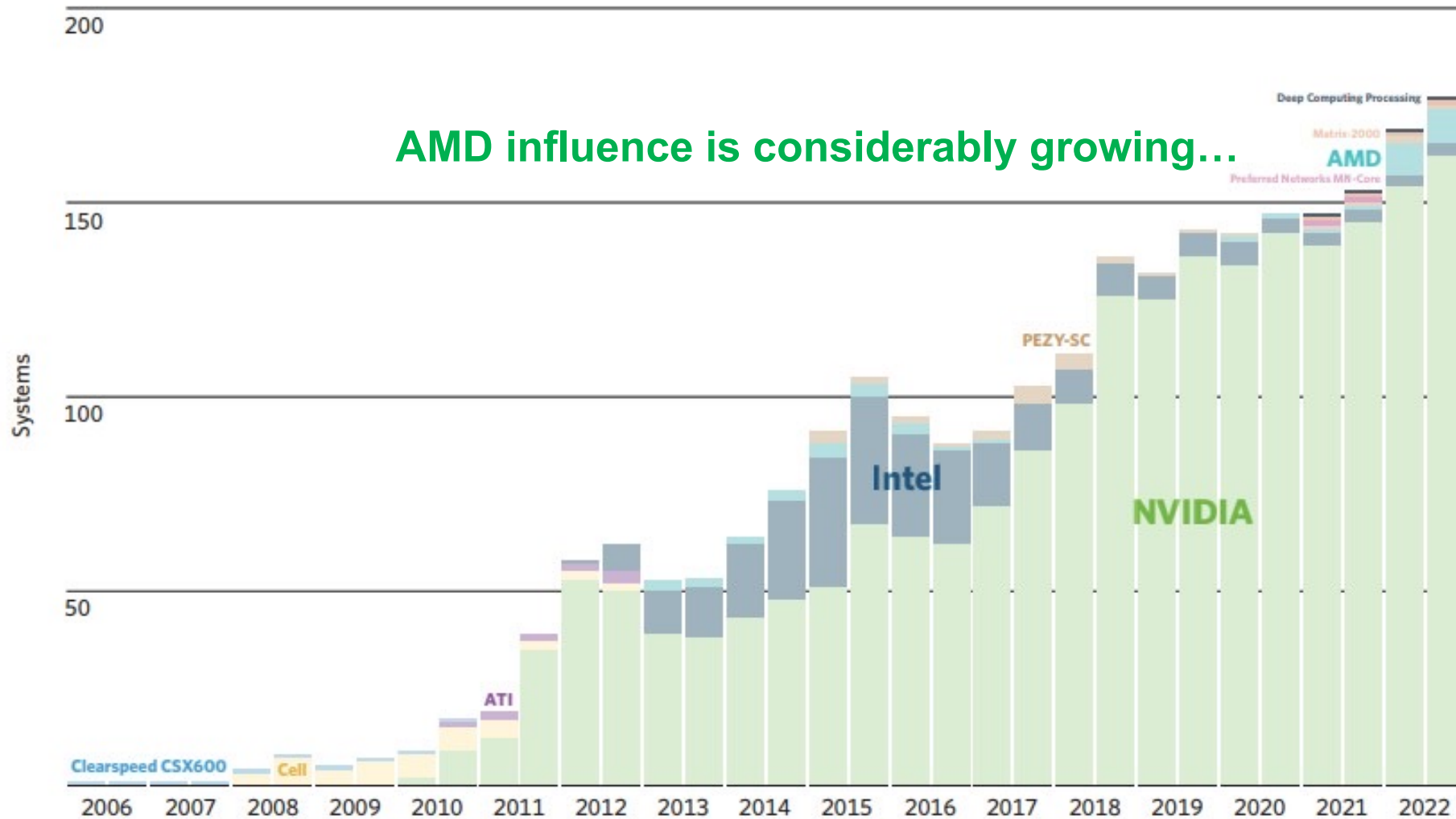


CoProcessor distribution

Nov'25



Accelerator families 2006 to 2022





1. TOP500

- a) TOP10 lists from Nov'17 to Nov'23
- b) Analysis of some relevant systems and architectures
- c) PU chip technology evolution in the past 25 years and since last year
- d) Evolution of the accelerators since they were available
- e) Country distribution over the past 25 years

2. GREEN500

- a) TOP5 list from Nov'19 to Nov'25
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3. HPCG

- a) HPCG vs. HPL: an overview
- b) TOP10 in Nov'22

4. HPL-AI

- a) High-performance Linpack (HPL) and artificial intelligence (AI) workloads

The Green500 list



The list ranks computers in terms of energy efficiency, typically measured as LINPACK FLOPS per watt.

About the Green500 List

The Green500 list ranks the top 500 supercomputers in the world by energy efficiency. The focus of performance-at-any-cost computer operations has led to the emergence of supercomputers that consume vast amounts of electrical power and produce so much heat that large cooling facilities must be constructed to ensure proper performance. To address this trend, the Green500 list puts a premium on energy-efficient performance for sustainable supercomputing.

The inaugural Green500 list was announced on November 15, 2007 at SC|07. As a complement to the TOP500, the unveiling of the Green500 ushered in a new era where supercomputers can be compared by performance-per-watt.

While the selection of any power-performance metric will be controversial, we currently opt for "FLOPS-per-Watt" given that it has already become a widely used metric in the community and for



TOP500		System	Cores	Rmax (PFlop/s)	Power (kW)	Energy Efficiency (GFlops/watts)
Rank	Rank					
1	420	KAIROS - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN CALMIP / University of Toulouse - CNRS France	13,056	3.05	46	73.282
2	171	ROMEO-2025 - BullSequana XH3000, Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, Red Hat Enterprise Linux, EVIDEN ROMEO HPC Center - Champagne-Ardenne France	47,328	9.86	160	70.912
3	225	Levante GPU extension - BullSequana XH3000, GH Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, RedHat Enterprise Linux, EVIDEN DKRZ - Deutsches Klimarechenzentrum Germany	35,904	6.75	110	69.426
4	213	Isambard-AI phase 1 - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE University of Bristol United Kingdom	34,272	7.42	117	68.835
5	286	Otus (GPU only) - ThinkSystem SD665-N V3, AMD EPYC 9655 96C 2.6GHz, NVIDIA H100 SXM5 80GB, Infiniband NDR, Rocky Linux 9.4, Lenovo Universitaet Paderborn - PC2 Germany	19,440	4.66		68.177
6	73	Capella - Lenovo ThinkSystem SD665-N V3, AMD EPYC 9334 32C 2.7GHz, Nvidia H100 SXM5 94Gb, Infiniband NDR200, AlmaLinux 9.4, MEGWARE TU Dresden, ZIH Germany	85,248	24.06	445	68.053

Top Green500 systems

Nov'25

Still Nvidia dominated...

7	334	SSC-24 Energy Module - HPE Cray XD670, Xeon Gold 6430 32C 2.1GHz, NVIDIA H100 SXM5 80GB, Infiniband NDR400, RHEL 9.2, HPE Samsung Electronics South Korea	11,200	3.82	69	67.251
8	96	Helios GPU - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cyfronet Poland	89,760	19.14	317	66.948
9	426	AMD Ouranos - BullSequana XH3000, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Infiniband NDR200, RedHat Enterprise Linux, EVIDEN Atos France	16,632	2.99	48	66.464
10	70	Portage - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, RHEL 8.9, HPE Hewlett Packard Enterprise United States	129,024	24.50	370	66.277



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HPCG benchmark



HPCG is a self-contained C++ program with MPI and OpenMP support that measures the performance of basic operations in a unified code:

HPCG benchmark

From Wikipedia, the free encyclopedia

- Sparse matrix-vector multiplication
- Vector updates
- Global dot products
- Local symmetric Gauss-Seidel smoother
- Sparse triangular solve (part of Gauss-Seidel smoother)

The **HPCG (high performance conjugate gradient) benchmark** is a [supercomputing benchmark](#) test proposed by Michael Heroux from [Sandia National Laboratories](#), and [Jack Dongarra](#) and Piotr Luszczek from the [University of Tennessee](#).^{[1][2]} It is intended to model the [data access](#) patterns of real-world [applications](#) such as [sparse matrix](#) calculations, thus testing the effect of limitations of the [memory subsystem](#) and internal [interconnect](#) of the supercomputer on its computing performance.^[3] Because it is internally [I/O bound](#), HPCG testing generally achieves only a tiny fraction of the peak [FLOPS](#) of the computer.^[4]

HPCG is intended to complement benchmarks such as the [LINPACK benchmarks](#) that put relatively little stress on the internal interconnect.^[5] The source of the HPCG benchmark is available on [GitHub](#).^[6]



TOP500: HPCG vs. HPL



HPCG Benchmark

The High Performance Conjugate Gradients (HPCG) Benchmark project is an effort to create a new metric for ranking HPC systems. HPCG is intended as a complement to the High Performance LINPACK (HPL) benchmark, currently used to rank the TOP500 computing systems. The computational and data access patterns of HPL are still representative of some important scalable applications, but not all. HPCG is designed to exercise computational and data access patterns that more closely match a different and broad set of important applications, and to give incentive to computer system designers to invest in capabilities that will have impact on the collective performance of these applications.

HPCG is a complete, stand-alone code that measures the performance of basic operations in a unified code:

- Sparse matrix-vector multiplication.
- Vector updates.
- Global dot products.
- Local symmetric Gauss-Seidel smoother.
- Sparse triangular solve (as part of the Gauss-Seidel smoother).
- Driven by multigrid preconditioned conjugate gradient algorithm that exercises the key kernels on a nested set of coarse grids.
- Reference implementation is written in C++ with MPI and OpenMP support.

<http://www.hpcg-benchmark.org>



Top Systems

Nov'25

NVIDIA
advantage?

Rank	TOP500 Rank	System	Cores	Rmax (PFlop/s)	HPCG (TFlop/s)
1	1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,340,000	1,809.00	17406.00
2	7	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	16004.50
3	2	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	14054.00
4	3	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	5612.60
5	9	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	4586.95
6	17	CHIE-4 - NVIDIA DGX B200, Xeon Platinum 8570 56C 2.1GHz, NVIDIA B200 SXM 180GB, Infiniband NDR400, Ubuntu 24.04.2 LTS, Nvidia SoftBank Corp. Japan	662,256	135.40	3760.55

7	8	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	2,121,600	434.90	3671.32
8	10	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA Italy	1,824,768	241.20	3113.94
9	15	Discovery 6 - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE ExxonMobil United States	806,208	164.20	2580.06
10	16	ABCI 3.0 - HPE Cray XD670, Xeon Platinum 8558 48C 2.1GHz, NVIDIA H200 SXM5 141 GB, Infiniband NDR200, Rocky Linux 9, HPE National Institute of Advanced Industrial Science and Technology (AIST) Japan	479,232	145.10	2445.67

European Supercomputing



European Supercomputing



EuroHPC
Joint Undertaking



European Supercomputing



EuroHPC
Joint Undertaking



NOV 2023	TOP500	Green500
LUMI	#5	#7
LEONARDO	#6	#18
MARENOSTRUM 5	#8	#6
MELUXINA	#71	#27
KAROLINA	#113	#25
DISCOVERER	#166	#216
VEGA	#198	#253



EuroHPC
Joint Undertaking

EuroHPC supercomputers: their locations

